THERMAL REVISION DECK

PART 1: CORE THERMODYNAMICS & COOLING EQUATIONS

CARD #01

THERMODYNAMICS

NEWTON'S LAW OF COOLING

Question: What fundamental equation governs the rate of heat loss of a hot core capsule submerged in a cooler ambient fluid?

SPHERE STUDY DECK

CARD #02

THERMODYNAMICS

DECAY CONSTANT (k)

Question: What does the constant k quantify physically, and which capsule is better insulated: $k = 0.15\ \mathrm{min}^{-1}$ or $k = 0.015\ \mathrm{min}^{-1}$?

SPHERE STUDY DECK

CARD #03

HEAT TRANSFER

FOURIER'S LAW OF CONDUCTION

Question: How does thermal energy transfer directly through solid matter, and what formula defines conduction heat flow?

SPHERE STUDY DECK

CARD #04

EQUILIBRIUM

THERMAL EQUILIBRIUM

Question: When does net heat transfer between the capsule core and the $0^\circ\mathrm{C}$ ice bath drop to zero? What physical law governs this?

SPHERE STUDY DECK

THERMAL REVISION DECK

PART 1: CORE THERMODYNAMICS & COOLING EQUATIONS

CARD #02

INSULATION FACTOR

LOWER-k CAPSULE ($k = 0.015\ \text{min}^{-1}$)

Answer: For a lumped system, $k = \frac{hA}{mc}$ is an effective cooling constant. Under comparable conditions, a smaller k means a slower approach to the ambient temperature.

Units: min⁻¹ or s⁻¹

CARD #01

KEY FORMULA

EXPONENTIAL DECAY

$$T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$$

Explanation: The cooling rate $\frac{dT}{dt}$ is directly proportional to the instantaneous temperature gap $(T - T_{\text{env}})$.

k = cooling constant

CARD #04

ZEROTH LAW

EQUAL TEMPERATURES

$$T_{\text{capsule}} = T_{\text{env}}$$

Answer: When $T_{\text{capsule}} = T_{\text{env}}$. For the nominal ice bath this limit is 0°C, but the measured bath temperature should be used. With no temperature difference, there is no net heat transfer between them.

Net heat-transfer rate = 0

CARD #03

CONDUCTION EQUATION

MOLECULAR VIBRATION

$$\dot{Q} = -\lambda A \frac{dT}{dx}$$

Explanation: The conductive heat-transfer rate depends on thermal conductivity λ, area A and the temperature gradient. Low-conductivity materials reduce the rate.

λ = thermal conductivity

THERMAL REVISION DECK

PART 2: RADIATIVE SHIELDING & MULTI-LAYER INSULATION

CARD #05

RADIATION

SPACE VACUUM RADIATION

Question: Which heat-transfer mechanism can cross a vacuum between separated bodies?

SPHERE STUDY DECK

CARD #06

SHIELDING

MYLAR RADIATIVE SHIELD

Question: How does a thin shiny aluminized Mylar foil protect spacecraft and astronauts from extreme solar thermal radiation?

SPHERE STUDY DECK

CARD #07

CONVECTION

CONVECTIVE AIR TRAPPING

Question: How do the sealed air cells in bubble wrap reduce heat transfer?

SPHERE STUDY DECK

CARD #08

MLI SPACESUITS

MULTI-LAYER INSULATION

Question: Why do spacesuits and satellites stack Cotton + Bubble Wrap + Mylar in composite layers?

SPHERE STUDY DECK

THERMAL REVISION DECK

PART 2: RADIATIVE SHIELDING & MULTI-LAYER INSULATION

CARD #06

OPTICAL REFLECTION

INFRARED REFLECTANCE

Answer: An aluminized reflective film can have high infrared reflectance and low emissivity. Its performance depends on the specific product, surface condition and whether it faces a suitable gap; it is not a substitute for conductive insulation in direct water contact.

Verify product optical properties

CARD #05

STEFAN-BOLTZMANN

PHOTON HEAT FLOW

$$P_{\text{rad}} = \epsilon\sigma A(T^4 - T_{\text{env}}^4)$$

Answer: Thermal radiation can cross a vacuum as electromagnetic energy. Conduction can still occur through physical contacts within a spacecraft, but not across an empty gap.

$\sigma = 5.67\times10^{-8}$ W/m²K⁴

CARD #08

TRIPLE SHIELD

TRIPLE HEAT SHIELD

Answer: Composite synergy:

- **Cotton:** Blocks solid conduction.
- **Bubble Wrap:** Traps fluid convection.
- **Mylar:** Reflects infrared radiation.

All 3 heat paths blocked

CARD #07

STAGNANT CELLS

STAGNANT AIR POCKETS

Answer: Air has low thermal conductivity ($\lambda_{\text{air}} \approx 0.026\ \mathrm{W/(m\cdot K)}$), while small sealed cells suppress bulk air circulation.

Low gas conduction + suppressed circulation

THERMAL REVISION DECK

PART 3: REFERENCE MODEL & EXPERIMENTAL BOUNDARY

CARD #09

SIMULATION

ANALYTICAL REFERENCE MODEL

Question: What assumptions does the SPHERE Newton cooling model make?

SPHERE STUDY DECK

CARD #10

LAB ANALOG

ICE-WATER BOUNDARY

Question: Why is an ice-water slurry used in the comparative experiment?

SPHERE STUDY DECK

CARD #11

THERMAL MASS

HEAT CAPACITY OF WATER

Question: Why is liquid water ($c = 4184\ \mathrm{J/kg} \cdot {}^{\circ}\mathrm{C}$) ideal for capsule core telemetry trials? What is the heat formula?

SPHERE STUDY DECK

CARD #12

GRAPH ANALYSIS

DECAY CURVE CURVATURE

Question: Why is the temperature-time graph $T(t)$ steep at $t = 0$ and increasingly flat toward $t = 15\ \mathrm{min}$?

SPHERE STUDY DECK

THERMAL REVISION DECK

PART 3: REFERENCE MODEL & EXPERIMENTAL BOUNDARY

CARD #10

CONSTANT SINK

CONSTANT HEAT SINK

Answer: The slurry provides a relatively stable, repeatable cold boundary. Its actual temperature must be measured; it does not reproduce vacuum heat transfer.

$T_{\text{env}} = 0.0°\text{C}$ constant

CARD #09

MODEL ASSUMPTIONS

LUMPED-PARAMETER MODEL

Answer: It assumes a uniform capsule temperature, a constant bath temperature and a constant effective cooling coefficient. The model is a reference calculation, not a full digital twin.

Hardware + Simulation

CARD #12

ASYMPTOTIC FLOW

SHRINKING TEMP GAP

Answer: Heat flux rate $\frac{dQ}{dt} \propto (T - T_{\text{env}})$. As the capsule cools, $(T - T_{\text{env}})$ shrinks, causing the rate of temperature drop to slow down asymptotically.

$\text{Slope} = -k(T - T_{\text{env}})$

CARD #11

THERMAL ENERGY

HIGH THERMAL MASS

$$Q = m \cdot c \cdot \Delta T$$

Answer: Water stores large thermal energy per gram, producing smooth, measurable exponential decay over $15+$ minutes without dropping instantly.

$c = 4184\ \text{J/kg} \cdot \text{K}$

THERMAL REVISION DECK

PART 4: METHOD, ERROR METRICS & SYNTHESIS

CARD #13

METRICS

MEAN ABSOLUTE ERROR

Question: How is the mean absolute error between measured and predicted temperatures calculated?

SPHERE STUDY DECK

CARD #14

LAB PROTOCOL

PROBE PLACEMENT

Question: Why should the thermometer probe be positioned near the centre of the liquid rather than touching the capsule wall?

SPHERE STUDY DECK

CARD #15

SI UNITS

SI UNITS REFERENCE

Question: What are the standard SI units for temperature T, thermal energy Q, thermal conductivity λ, and cooling constant k?

SPHERE STUDY DECK

CARD #16

DESIGN COMPARISON

MULTILAYER CONFIGURATION

Question: Which tested configuration is assigned the smallest reference cooling constant, and which mechanisms may explain that choice?

SPHERE STUDY DECK

THERMAL REVISION DECK

PART 4: METHOD, ERROR METRICS & SYNTHESIS

CARD #14

BOUNDARY LAYER

BULK FLUID TEMPERATURE

Answer: Wall contact biases the reading towards the local boundary temperature rather than the bulk liquid temperature and introduces a systematic measurement error.

Use consistent probe position

CARD #13

MODEL ACCURACY

MODEL RESIDUALS

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|T_{\mathrm{meas},i} - T_{\mathrm{model},i}\right|$$

Answer: MAE is the mean magnitude of the residuals and is reported in $^\circ\mathrm{C}$. A lower value indicates closer agreement but does not, by itself, validate every model assumption.

Units: °C

CARD #16

REFERENCE RESULT

MULTILAYER TEST ASSEMBLY

Answer: The cotton, bubble-wrap and reflective-film assembly is assigned $k = 0.015\ \mathrm{min}^{-1}$. This is a model assumption to be evaluated against measurements, not a guaranteed material property.

Compare with measured data

CARD #15

STANDARD UNITS

UNITS CHEAT SHEET

- **Temperature (T):** Kelvin (K) or Celsius $(^\circ\mathrm{C})$
- **Thermal Energy (Q):** Joules (J)
- **Decay Constant (k):** s^{-1} or min^{-1}
- **Conductivity (λ):** $\mathrm{W}/(\mathrm{m}\cdot\mathrm{K})$

Standard Metric Units